

Sanando Banerjee

| [Email](#) | +916289941394 | [Linkedin](#) | [GitHub](#) | [Portfolio](#) |

PROFESSIONAL SUMMARY

Software Engineering undergraduate focused on backend and AI systems, with hands-on experience building API-driven, real-time, and codebase-aware AI applications. Experienced in designing modular backend services, retrieval pipelines, data processing workflows, and full-stack systems using Python and modern web technologies.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, REST APIs, Server-Sent Events

Frontend: React, Next.js

Databases: MongoDB, ChromaDB, SQL Server

AI/ML: LangChain, LangGraph, RAG, MCP

Tools & Testing: Git, Docker, Linux, Playwright

PROJECT EXPERIENCE

Saturn-AI – Real-Time AI Trading Intelligence Platform

- **Engineered** a real-time Bitcoin intelligence platform that ingests news from **5 RSS sources every 10 minutes**, performs AI-powered sentiment analysis, aggregated into **BUY/SELL/HOLD signals with confidence scores**.
- **Designed** a modular FastAPI backend with **10 REST endpoints**, MongoDB persistence, scheduled processing via APScheduler, and a streaming AI explanation service for real-time market analysis.
- **Integrated** a React dashboard, CoinGecko price data, and an **MCP server** for AI-agent access; added **Playwright E2E tests** and **GitHub Actions CI/CD** for automated validation.

Mars-AI – Multi-hop Agentic Retrieval and Scoring

- Engineered a codebase-aware coding agent that combines **AST-based Python parsing**, **semantic retrieval**, and **3-hop call-graph traversal** to answer repository-level questions with grounded source citations.
- Designed a **LangGraph retrieval pipeline** using ChromaDB and local **sentence-transformer embeddings**, dynamically deciding when additional caller/callee traversal is required to resolve a query.
- Built a **FastAPI backend** and **Next.js interface** with **citation metadata** generated directly from parsed code, preventing LLM-generated source references and grounding responses in retrieved code context.

RELEVANT EXPERIENCE

bITsera – Tech Club of IT, MSIT | Web Development Lead

Led and organized React workshops for 60+ students, covering state management, API integration and React hooks while emphasizing the importance of modular architecture and reusable component design.

EDUCATION

Meghnad Saha Institute of Technology

Bachelor of Technology (B. Tech) in Information Technology

Kolkata, India

Aug 2023 - Jul 2027
